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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Fourier Transform

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

38	27	1992 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	19 questions
Marks 2	16 questions
Marks 5	3 questions
Answer key	at the end

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Marks 1 19 questions

Q1. GATE ECE 2025

Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t)\exp(-j\omega t)dt$ exists.

Which one of the following statements is always TRUE?

(A) $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|dt$

(B) $|F(\omega)| > \int_{-\infty}^{\infty} |f(t)|dt$

(C) $|F(\omega)| \leq \int_{-\infty}^{\infty} f(t)dt$

(D) $|F(\omega)| \geq \int_{-\infty}^{\infty} f(t)dt$

Open solution

Q2. GATE ECE 2023

Let $m(t)$ be a strictly band-limited signal with bandwidth B and energy E. Assuming $\omega_0 = 10B$, the energy in the signal $m(t)\cos\omega_0 t$ is

(A) $\frac{E}{4}$ (B) $\frac{E}{2}$ (C) E (D) 2E

Open solution

Q3. GATE ECE 2023

The Fourier transform $x(\omega)$ of $x(t) = e^{-t^2}$ is

Note : $\int_{-\infty}^{\infty} e^{-y^2}dy = \sqrt{\pi}$

(A) $\sqrt{\pi}e^{\frac{\omega^2}{2}}$

(B) $\frac{e^{-\frac{\omega^2}{4}}}{2\sqrt{\pi}}$

(C) $\sqrt{\pi}e^{-\frac{\omega^2}{4}}$

(D) $\sqrt{\pi}e^{-\frac{\omega^2}{2}}$

Open solution

Q4. GATE ECE 2021

Consider a real-valued base-band signal $x(t)$, band limited to 10 kHz . The Nyquist rate for the signal $y(t) = x(t)\,x\left(1+\frac{t}{2}\right)$ is

(A) 20 kHz (B) 30 kHz (C) 60 kHz (D) 15 kHz

Open solution

Q5. GATE ECE 2016 Set 2

The energy of the signal x(t) = $\frac{\sin(4\pi t)}{4\pi t}$ is _____.

Open solution

Q6. GATE ECE 2016 Set 1

Which one of the following is an eigenfunction of the class of all continuous-time, linear, time-invariant systems (u(t) denotes the unit-step function)?

(A) $e^{j\omega_0 t}u(t)$ (B) $\cos(\omega_0 t)$ (C) $e^{j\omega_0 t}$ (D) $\sin(\omega_0 t)$

Open solution

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Q7. GATE ECE 2016 Set 3

If the signal x(t) = $\frac{\sin(t)}{\pi t} * \frac{\sin(t)}{\pi t}$ with * denoting the convolution operation, then x(t) is equal to

(A) $\frac{\sin(t)}{\pi t}$
(B) $\frac{\sin(2t)}{\pi t}$
(C) $\frac{2\sin(t)}{\pi t}$
(D) $\left(\frac{\sin(t)}{\pi t}\right)^2$

Open solution

Q8. GATE ECE 2006

Let x(t) $\leftrightarrow$ X($(j\omega)$ BE Fourier transform pair. The Fourier Transform of the signal x(5t - 3) in terms of X($(j\omega)$ is given as

(A) $\frac{1}{5}e^{-\frac{j3\omega}{5}}X\left(\frac{j\omega}{5}\right)$
(B) $\frac{1}{5}e^{\frac{j3\omega}{5}}X\left(\frac{j\omega}{5}\right)$
(C) $\frac{1}{5}e^{-j3\omega}X\left(\frac{j\omega}{5}\right)$
(D) $\frac{1}{5}e^{j3\omega}X\left(\frac{j\omega}{5}\right)$

Open solution

Q9. GATE ECE 2004

The Fourier transform of a conjugate symmetric function is always

(A) imaginary
(B) conjugate anti-symmetric
(C) real
(D) conjugate symmetric

Open solution

Q10. GATE ECE 2002

The Fourier transform F $\{e^{-t}u(t)\}$ is equal to $\frac{1}{1+j2\pi f}$. Therefore, $F\left\{\frac{1}{1+j2\pi t}\right\}$ is

(A) $e^{f}u(f)$
(B) e^{-f} u(f)
(C) $e^{f}u(-f)$
(D) e^{-f}u(-f)

Open solution

Q11. GATE ECE 2001

If a signal f(t) has energy E, the energy of the signal f(2t) is equal to

(A) E
(B) E/2
(C) 2E
(D) 4E

Open solution

Q12. GATE ECE 2000

The Fourier Transform of the signal $x(t) = e^{-3t^2}$ is of the following form, where A and B are constants:

(A) $Ae^{-B|f|}$
(B) Ae^{-Bf}
(C) $A + B|f|^2$
(D) Ae^{-Bf^2}

Open solution

Q13. GATE ECE 1999

A signal x(t) has a Fourier transform X (ω). If x(t) is a real and odd function of t, then X(ω) is

(A) a real and even function of ω
(B) an imaginary and odd function of ω
(C) an imaginary and even function of ω
(D) a real and odd function of ω

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Q14. GATE ECE 1999

A modulated signal is given by s(t)= e^{-at} cos $[(\omega_c + \Delta\omega)t]$ u (t), where a, ω_c and $\Delta\omega$ are positive constants, and ω_c >>$\Delta\omega$. The complex envelope of s(t) is given by

(A) exp(-at)exp$[(\omega_c + \Delta\omega)t]$ u(t)
(B) exp (-at)exp(j$\Delta\omega t$) u(t)
(C) exp(j$\Delta\omega t$) u (t)
(D) exp$[j(\omega_c + \Delta\omega)t]$

Open solution

Q15. GATE ECE 1998

The amplitude spectrum of a Gaussian pulse is

(A) uniform
(B) a sine function
(C) Gaussian
(D) an impulse function

Open solution

Q16. GATE ECE 1998

The Fourier transform of a function x(t) is X(f). The Fourier transform of $\frac{dx(t)}{dt}$ will be

(A) $\frac{dX(f)}{dt}$
(B) $j2\pi fX(f)$
(C) $jfX(f)$
(D) $\frac{X(f)}{jf}$

Open solution

Q17. GATE ECE 1998

The Fourier transform of a voltage of a voltage signal x(t) is X(f). The unit of |X(f)| is

(A) Volt
(B) Volt - sec
(C) Volt/sec
(D) $Volt^2$

Open solution

Q18. GATE ECE 1997

The function f(t) has the Fourier Transform g(ω). The Fourier Transform of $g(t) = \left(\int_{-\infty}^{\infty} g(t)e^{-j\omega t}dt\right)$ is

(A) $\frac{1}{2\pi}f(\omega)$
(B) $\frac{1}{2\pi}f(-\omega)$
(C) $2\pi\, f(-\omega)$
(D) none of the above

Open solution

Q19. GATE ECE 1996

The Fourier transform of a real valued time signal has

(A) odd symmetry
(B) even symmetry
(C) conjugate symmetry
(D) no symmetry

Open solution

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Marks 2 16 questions

Q20. GATE ECE 2026

Open solution

Consider a real baseband signal $x(t) = e^{-2t}$, for t (in seconds) ≥ 0. If 99% of energy of $x(t)$ lies within B Hz, then which of the following options is TRUE for the value of B ?

(A) $B > 1\text{kHz}$

(B) \$63 / \pi \mathrm{Hz}

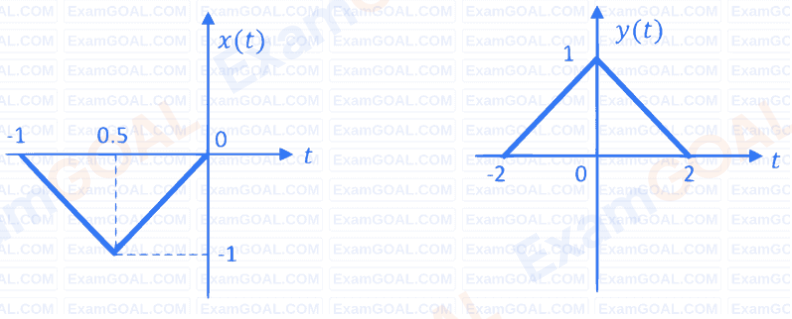
(C) $126/\pi\text{Hz} < \text{B} < 128/\pi\text{Hz}$

(D) $B < 1$ Hz

Q21. GATE ECE 2024

Open solution

Consider two continuous time signals $x(t)$ and $y(t)$ as shown below



If $X(f)$ denotes the Fourier transform of $x(t)$, then the Fourier transform of $y(t)$ is _____.

(A) $-4X(4f)e^{-j\pi f}$

(B) $-4X(4f)e^{-j4\pi f}$

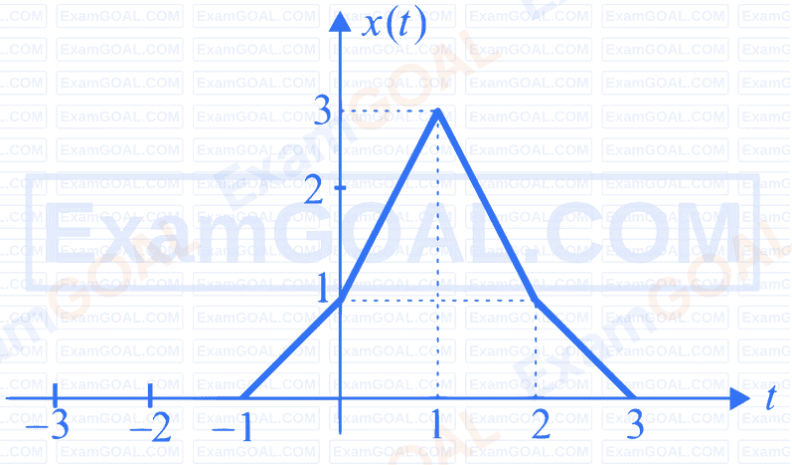
(C) $-\frac{1}{4}X(f/4)e^{-j\pi f}$

(D) $-\frac{1}{4}X(f/4)e^{-j4\pi f}$

Q22. GATE ECE 2020

Open solution

$X(\omega)$ is the Fourier transform of $x(t)$ shown below. The value of $\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$ (rounded off to two decimal places) is ____ .



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Q23. GATE ECE 2015 Set 3

The complex envelope of the bandpass signal $x(t) = -\sqrt{2}\left(\frac{\sin(\pi t/5)}{\pi t/5}\right)\sin\left(\pi t - \frac{\pi}{4}\right)$, centered about f = $\frac{1}{2}\,Hz$, is

(A) $\left(\frac{\sin(\pi t/5)}{\pi t/5}\right)e^{j\frac{\pi}{4}}$

(B) $\left(\frac{\sin(\pi t/5)}{\pi t/5}\right)e^{-j\frac{\pi}{4}}$

(C) $\sqrt{2}\left(\frac{\sin(\pi t/5)}{\pi t/5}\right)e^{j\frac{\pi}{4}}$

(D) $\sqrt{2}\left(\frac{\sin(\pi t/5)}{\pi t/5}\right)e^{-j\frac{\pi}{4}}$

Open solution

Q24. GATE ECE 2015 Set 2

The value of the integral $\int_{-\infty}^{\infty} 12\,\cos(2\pi t)\,\frac{\sin(4\pi t)}{4\pi t}\,dt$ is

Open solution

Q25. GATE ECE 2014 Set 1

For a function g(t), it is given that $\int_{-\infty}^{\infty} g(t)e^{-j\omega t}dt = \omega e^{-2\omega^2}$ for any real value ω. If y(t)=$\int_{-\infty}^{t} g(\tau)d\tau$, $then\ \int_{-\infty}^{\infty} y(t)\,dt$ is

(A) 0 (B) - j (C) $-\frac{j}{2}$ (D) $\frac{j}{2}$

Open solution

Q26. GATE ECE 2014 Set 2

The value of the integral $\int_{-\infty}^{\infty} \mathrm{sinc}^2(5t)\,dt$ is

Open solution

Q27. GATE ECE 2012

The Fourier transform of a signal h(t) is $H(j\omega)$ =(2 cos ω) (sin 2ω) / ω. The value of h(0) is

(A) 1/4 (B) 1/2 (C) 1 (D) 2

Open solution

Q28. GATE ECE 2008

The signal x(t) is described by $x(t) = \begin{cases} 1 & for\ -1 \le t \le +1 \\ 0 & otherwise \end{cases}$

Two of the angular frequencies at which its Fourier transform becomes zero are

(A) $\pi,\ 2\pi$ (B) $0.5\,\pi,\ 1.5\,\pi$ (C) $0,\ \pi$ (D) $2\,\pi,\ 2.5\,\pi$

Open solution

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Q29. GATE ECE 2005

For a signal x(t) the Fourier transform is X(f). Then the inverse Fourier transform of X(3f+2) is given by

(A) $\frac{1}{2}\,x\left(\frac{t}{2}\right)e^{j3\pi t}$

(B) $\frac{1}{3}\,x\left(\frac{t}{3}\right)e^{-j4\pi t/3}$

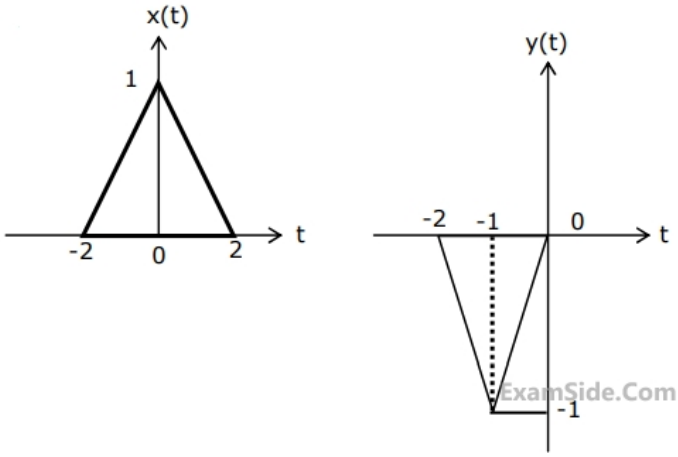
(C) $3\,x(3t)e^{-j4\pi t}$

(D) $x(3t+2)$

Open solution

Q30. GATE ECE 2004

Let x(t) and y(t) (with Fourier transforms X(f) and Y(f) respectively) be related as shown in Fig.(1) & (2).



Then Y(f) is

(A) $-\frac{1}{2}X(f/2)e^{-j2\pi f}$

(B) $-\frac{1}{2}X(f/2)e^{j2\pi f}$

(C) $-X(f/2)e^{j2\pi f}$

(D) $-X(f/2)e^{-j2\pi f}$

Open solution

Q31. GATE ECE 2000

The Hilbert transform of $[\cos\,\omega_1 t + \sin\omega_2 t\,]$ is

(A) $\sin\,\omega_1 t - \cos\omega_2 t$

(B) $\sin\,\omega_1 t + \cos\omega_2 t$

(C) $\cos\,\omega_1 t + \sin\omega_2 t$

(D) $\sin\omega_1 t + \sin\omega_2 t$

Open solution

Q32. GATE ECE 1997

The power spectral density of a deterministic signal is given by $[\sin(f)/f]^2$, where 'f' is frequency. The autocorrelation function of this signal in the time domain is

(A) a rectangular pulse

(B) a delta function

(C) a sine pulse

(D) a triangular pulse

Open solution

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Q33. GATE ECE 1997

If the Fourier Transfrom of a deterministic signal g(t) is G (f), then

Item-1

(1) The Fourier transform of g (t - 2) is

(2) The Fourier transform of g (t/2) is

Item - 2

(A) G(f) $e^{-j(4\pi f)}$

(B) G(2f)

(C) 2G(2f)

(D) G(f-2)

Match each of the items 1, 2 on the left with the most appropriate item A, B, C or D on the right.

(A) (1 - A), (2 - C) **(B)** (I - B), (2 - C) **(C)** (1 - D), (2 - C) **(D)** (1 - A), (2 - D)

Open solution

Q34. GATE ECE 1996

The autocorrelation function of an energy signal has

(A) no symmetry **(B)** conjugate symmetry

(C) odd symmetry **(D)** even symmetry

Open solution

Q35. GATE ECE 1992

If G(f) represents the Fourier transform of a signal g (t) which is real and odd symmetric in time, then

(A) G(t) is complex **(B)** G(t) is imaginary

(C) G(t) is real **(D)** G(t) is real and non- negative

Open solution

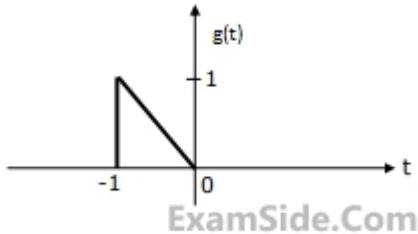
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Marks 5 3 questions

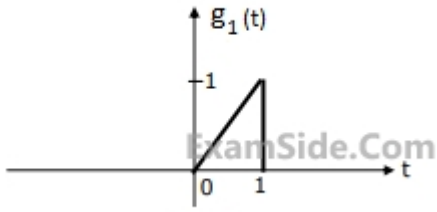
Q36. GATE ECE 2001

Open solution

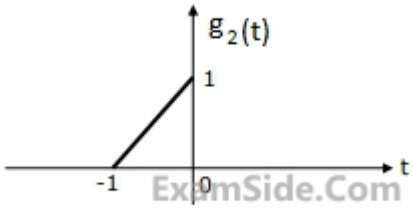
The Fourier transform $G(\omega)$ of the signal g(t) in Fig.(1) is given as
$G(\omega) = \frac{1}{\omega^2}(e^{j\omega} - j\omega e^{j\omega} - 1)$.
Using this information, and the time-shifting and time-scaling properties, determine and Fourier transform of signals in Fig (2), (3) and (4).



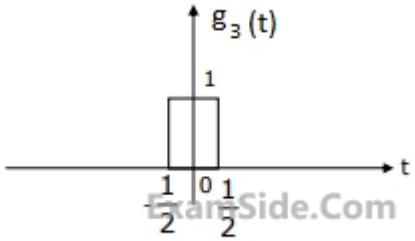
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Fig.(1)



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Fig.(2)



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Fig. (3)



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Fig.(4)

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Q37. GATE ECE 1998

Consider a rectangular pulse g(t) existing between $t = -\frac{T}{2} and \frac{T}{2}$. Find and sketch the pulse obtained by convolving g(t) with itself. The Fourier transform of g(t) is a sinc function. Write down the Fourier transform of the pulse obtained by the above convolution.

Open solution

Q38. GATE ECE 1995

A signal v(t)= [1+ m(t)] cos $(\omega_c t)$ is detected using a square law detector, having the characteristic $v_0(t) = v^2(t)$. If the Fourier transform of m(t) is constant, M_0, extending from - f_m to $+ f_m$, sketch the Fourier transform of $v_0(t)$ in the frequency range-f_m $< f < f_m$.

Open solution

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Answer Key

Marks 1

1. **A**	2. **B**	3. **C**	4. **B**	5. **0.25**
6. **C**	7. **A**	8. **A**	9. **C**	10. **C**
11. **B**	12. **D**	13. **B**	14. **B**	15. **C**
16. **B**	17. **B**	18. **C**	19. **C**	

Marks 2

20. **B**	21. **A**	22. **58.1 to 59.1**	23. **C**	24. **0.2**
25. **B**	26. **0.2**	27. **C**	28. **A**	29. **B**
30. **B**	31. **A**	32. **D**	33. **A**	34. **B**
35. **B**				

Marks 5

36. $G_1(\omega) = \frac{1}{\omega^2}\left(e^{-j\omega} + j\omega e^{-j\omega} - 1\right)$ $G_2(\omega) = \frac{1}{\omega^2}\left(1 + j\omega - e^{j\omega}\right)$ $G_3(\omega) = e^{j0.5\omega} G_1(\omega) + e^{-j0.5\omega} G(\omega)$	37. **Solve it.**	38. **Solve it.**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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